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LASER LEVEL

Abstract

A laser level including a control mechanism housing which houses a control mechanism, the control mechanism including at least a top surface and a bottom surface. A protective structure extends from the control mechanism housing configured to protect the control mechanism from impact and including an upper structure which extends from a top surface of the control mechanism housing and a lower structure which extends from a bottom surface of the control mechanism housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/440,433 filed Feb. 13, 2024 which is a continuation of U.S. application Ser. No. 18/383,276 filed Oct. 24, 2023, now U.S. Pat. No. 12,253,359, which is a continuation of U.S. application Ser. No. 17/874,380 filed Jul. 27, 2022, now U.S. Pat. No. 11,859,975, which is a continuation of U.S. application Ser. No. 16/951,076 filed Nov. 18, 2020, now U.S. Pat. No. 11,428,529, which is a continuation of U.S. application Ser. No. 16/589,411 filed Oct. 1, 2019, now U.S. Pat. No. 10,871,372, which is a continuation of U.S. application Ser. No. 16/038,617 filed Jul. 18, 2018, now U.S. Pat. No. 10,466,049, which is a continuation of U.S. application Ser. No. 14/936,047 filed Nov. 9, 2015, now U.S. Pat. No. 10,066,939, which claims the benefit of U.S. Provisional Application No. 62/240,720 filed on Oct. 13, 2015. The entire disclosures of each of the above is incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates to laser levels, particularly laser levels with a protective structure.

BACKGROUND

[0003] There are various existing laser levels. It is desired to provide a laser level with an improved protective structure.

SUMMARY

[0004] According to an exemplary embodiment, there is a laser level. The laser level includes a control mechanism housing which houses a control mechanism. A protective structure extends from the control mechanism housing and is configured to protect the control mechanism from impact.

[0005] The laser level may include a laser projector.

[0006] The control mechanism housing may include a top surface and wherein the laser projector is disposed at the top surface of the control mechanism housing.

[0007] The protective structure may include an upper structure which extends from the top surface of the control mechanism housing and extends beyond the laser projector.

[0008] The upper structure may extend both upwardly and outwardly from the top surface of the control mechanism housing.

[0009] The upper structure comprises an upper portion and a plurality of legs projecting from the control mechanism housing to the upper portion.

[0010] The control mechanism housing is cube shaped and the legs are disposed at corners of the top surface of the control mechanism housing.

[0011] A top surface of the upper structure may have a substantially square shape.

[0012] A top surface of the upper structure may be a closed shape.

[0013] The top surface of the upper structure may be open inside the closed shape.

[0014] The control mechanism housing may include a bottom surface opposite the top surface.

[0015] The protective structure may be made of a shock absorbing material.

[0016] The shock absorbing material may include at least one of rubber, foam and a shock absorbing plastic.

[0017] The control mechanism may include at least one laser.

[0018] The control mechanism may include a motor.

[0019] The control mechanism may include at least one lens.

[0020] The protective structure may further include a lower structure which extends from a bottom surface of the control mechanism housing.

[0021] The lower structure may extend both downwardly and outwardly from the top surface of the control mechanism housing.

[0022] The lower structure may include a lower portion and a plurality of lower legs projecting from the control mechanism housing to the lower portion.

[0023] The control mechanism housing may be cube shaped and the lower legs may be disposed at corners of the bottom surface of the control mechanism housing. A bottom surface of the lower structure may have a substantially square shape.

[0024] A bottom surface of the lower structure may be a closed shape.

[0025] The bottom surface of the lower structure may be open inside the closed shape.

[0026] The laser level may further include a battery to power the laser level.

[0027] The battery may be a removable battery pack.

[0028] The removable battery pack may be configured to power a variety of power tools and other products including such products as a drill, saw, sander, radio, infrared detector, lawn mower, string trimmer.

[0029] The laser level may be configured such that when placed on a flat surface, only the protective structure contacts the flat surface regardless of the orientation of the laser level.

[0030] The laser level may be configured such that it can be stably positioned on a flat surface in at least six orientations.

[0031] The at least six orientations may correspond to the six sides of the control mechanism housing.

[0032] There may be spaces between the control mechanism housing and the flanges.

[0033] The flanges may be usable as handles.

[0034] The upper protective structure may be made of rubber.

[0035] The legs of the upper protective structure may be made of rubber with a first durometer and the flanges of the upper protective structure may be made of a rubber with a second durometer, different than the first durometer.

[0036] The first durometer may be higher than the first durometer.

[0037] The second durometer may be lower than the first durometer.

[0038] According to another exemplary embodiment, the flanges of the upper protective structure may be made of rubber and the legs of the upper protective structure may be made of a material other than rubber. The material other than rubber may be metal. The metal may be spring steel.

[0039] The control mechanism housing may be made of a rigid material. The rigid material may be acrylonitrile butadiene styrene (ABS). The rigid material may be high impact polypropylene. The rigid material may be high impact polystyrene.

[0040] According to another exemplary embodiment, the flanges could form a generally elliptical shape and the control mechanism housing could be a generally elliptical cylinder.

[0041] According to another exemplary embodiment, the flanges of the protective structure can form a triangular shape and the control mechanism housing can be a generally triangular prism

shape. In other embodiments, the flanges could form a shape with five sides, six sides, seven sides, eight sides or more and the control mechanism housing can be shaped with a corresponding structure. In other embodiments, the shape formed by the flanges and the control mechanism housing may not correspond. For example, the flanges may form a hexagon shape while the control mechanism housing is generally cube shaped.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0042] FIG. 1 illustrates a perspective view of an exemplary embodiment of a laser level;
[0043] FIG. 2 is another perspective view of the exemplary embodiment of the laser level;
[0044] FIG. 3 is a top view of the exemplary embodiment of the laser level;
[0045] FIG. 4 is a back view of the exemplary embodiment of the laser level showing the attached battery pack;
[0046] FIG. 5 is a left side view of the exemplary embodiment of the laser level;
[0047] FIG. 6 is a right side view of the exemplary embodiment of the laser level;
[0048] FIG. 7 is a front view of the exemplary embodiment of the laser level;
[0049] FIG. 8 is a bottom view of the exemplary embodiment of the laser level;
[0050] FIG. 9 is perspective view of the exemplary embodiment of the laser level with the battery pack removed;
[0051] FIG. 10 is a top view of the exemplary embodiment of the laser level with the battery pack removed;
[0052] FIG. 11 is a back view of the exemplary embodiment of the laser level showing the battery pack connection section with the battery pack removed;
[0053] FIG. 12 is a left side view of the exemplary embodiment of the laser level with the battery pack removed;
[0054] FIG. 13 is a right side view of the exemplary embodiment of the laser level with the battery pack removed;
[0055] FIG. 14 is a front view of the exemplary embodiment of the laser level with the battery pack removed;
[0056] FIG. 15 is a bottom view of the exemplary embodiment of the laser level with the battery pack removed;
[0057] FIG. 16 is an illustration of a laser level according to a second exemplary embodiment;
[0058] FIG. 17 is an illustration of a laser level according to a third exemplary embodiment;
[0059] FIG. 18A is a simplified electrical schematic according to an exemplary embodiment; and
[0060] FIG. 18B is a simplified schematic according to an exemplary embodiment.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0061] An exemplary embodiment according to the present application is shown in FIGS. 1-15. As shown in FIG. 1, the exemplary embodiment shows a rotary laser level 10. Rotary laser levels are known, for example, as shown in U.S. Pat. Nos. 4,854,703; 4,751,782; and 6,338,681, which are herein incorporated by reference in their entirety. Another rotary laser level is shown in US Patent Application Publication No. 2014/0203172, which is hereby incorporated by reference. The present application may also be applicable to other types of lasers such as U.S. Pat. Nos. 7,665,217; 7,076,880; 6,964,106; 7,481,002; 7,027,480; 8,640,350; 6,606,798; 7,013,571; 7,111,406; 7,296,360; and 7,571,546, which are herein incorporated by reference in their entirety.

[0062] FIGS. 1-8 illustrate the exemplary embodiment of the invention with the removable battery pack 200 attached. FIGS. 9-15 illustrate the exemplary embodiment of the invention without the battery pack.

[0063] As shown in the Figs., there is a laser level 10. The laser level 10 includes a control

mechanism housing **20**. A laser projector **30** extends from the control mechanism housing and is configured to project a laser onto a surface. In the case of the rotary laser shown, the laser projection can be a 360 degree rotary projection. In other instances, the projector **30** may project one or more dot, one or more lines or a combination of lines and dots.

[0064] The control mechanism housing **20** includes a control mechanism which provides for projection of one or more laser beams or dots by the laser projector **30**. The control mechanism may include, among other things; an LED or other light source; one or more lenses; one or more mirrors; a motor; and a microprocessor configured to control the laser level **10**. The control mechanism may be a control mechanism shown in one of U.S. Pat. Nos. 4,854,703; 4,751,782; 6,338,681; US 2014/0203172; U.S. Pat. Nos. 7,665,217; 7,076,880; 6,964,106; 7,481,002; 7,027,480; 8,640,350; 6,606,798; 7,013,571; 7,111,406; 7,296,360; and 7,571,546 all of which have been incorporated by reference. For example, the control mechanism housing **20** may include the control mechanism housed in the upper casing part shown and described U.S. Pat. No. 4,854,703. Alternatively, the control mechanism housing **20** may include the control mechanism shown and described U.S. Pat. No. 8,640,350. In various embodiments, the projector **30** may be disposed at different places along the control mechanism housing **20**. For example, the projector **30** may be on a front surface or may be internal to the control mechanism housing **20** with beams projecting out from the control mechanism housing **20**.

[0065] The control mechanism housing **20** has a substantially cubical shape. Accordingly, it has a top surface **21**, a bottom surface **22**, a left surface **23**, a right surface **24**, a front surface **25** and a back surface **26**. The removable battery pack **200** is provided at the back surface. The removable battery pack **200** provides power for the laser level **10**. The removable battery pack **200** may be a power tool battery pack such that it can be removed and mounted to a variety of power tools, outdoor power tools, cleaning tools, or other tools or products. As shown in FIG. **11**, the laser level **10** includes a receptacle **201** for receiving the battery pack. The receptacle **201** can be one of many known designs for receiving a battery pack, including those for receiving a power tool battery pack. The control mechanism housing **20** may be made of a rigid material such as Acrylonitrile butadiene styrene (ABS), high impact polypropylene or high impact polystyrene (HIPS). In an embodiment, the control mechanism housing **20** may be made of a material having a Rockwell R hardness of 60 to 140. In other embodiments, the control mechanism housing may be made of a material having a Rockwell R hardness of 70 to 130, 80 to 120 or 80 to 114.

[0066] As shown in FIGS. **1-15**, the laser level **10** has a protective structure **50**. The protective structure **50** comprises an upper protective structure **100** and a lower protective structure **150**. The upper protective structure **100** projects upwardly and outwardly from the top surface **21** and the lower protective structure **150** projects downwardly and outwardly from the bottom surface **22**. The protective structure **50** in the exemplary embodiment is made of a shock absorbing material which can deform on impact at a controlled rate to dissipate the impact energy over a longer period. In the exemplary embodiment, the material of the protective structure **50** is designed to absorb shocks better than the material of the control mechanism housing **20**. The shock absorbing material may be a material such as rubber, foam or a shock absorbing plastic. As can be appreciated, because the protective structure extends beyond the control mechanism housing **20**, it protects the control mechanism housing **20** from impact when dropped from a variety of orientations. The flanges and the legs may be made of different durometers of rubber. For example, the rubber used for one or more legs may have a higher durometer than the rubber used for one or more flange. The rubber used for one or more flange may have a higher durometer than one or more legs.

[0067] In other exemplary embodiments, it is contemplated that the legs may be made from metal, such as spring steel, for example. In other exemplary embodiments, the protective structure may comprise a molded skeleton made from an impact resistant polymer that is overmolded in rubber or foamed rubber.

[0068] The upper protective structure **100** has four flanges **101**, **102**, **103**, and **104**. The flanges run

roughly parallel to upper edges of the control mechanism housing **20**. The flanges **101**, **102**, **103** and **104** are connected to the control mechanism housing **20** by upper legs **110**. The lower protective structure **150** has four flanges **151**, **152**, **153**, and **154**. The flanges run roughly parallel to upper edges of the control mechanism housing **20**. The flanges **151**, **152**, **153** and **154** are connected to the control mechanism housing **20** by lower legs **160**.

[0069] In the exemplary embodiment, each upper corner includes a pair of legs **110** and each lower corner includes a pair of legs **160**. In other embodiments, there may be additional or fewer legs. For example, each corner may include only one leg. The legs can also be dimensioned differently. For example, they could be made thinner or thicker than shown in the exemplary embodiment. In the exemplary embodiment, the legs **110**, **160** are made of the same material as the flanges **101**, **102**, **103**, **104**, **151**, **152**, **153**, and **154**. In other embodiments, the legs **110**, **160** may be made of a different material than the flanges. The legs **110**, **160** may be the same or differ in various ways.

[0070] The upper protective structure **100** is configured so that the flanges **101**, **102**, **103** and **104** rise above the projector **30**. In this way, the projector **30** is particularly protected against impact and the flanges **101**, **102**, **103** and **104** do not block lasers projecting from the projector **30**. In the shown exemplary embodiment, the upper legs **110** are longer than the lower legs **160**. This allows the upper legs **110** to provide sufficient clearance for the projector **30** so that the flanges **101**, **102**, **103** and **104** do not block any projection from the projector **30**.

[0071] The legs creates spaces **111**, **112**, **113**, **114**, **161**, **162**, **163** and **163** between the control mechanism housing **20** and the flanges **101**, **102**, **103**, **104**, **151**, **152**, **153**, and **154**. The spaces allow for a decreased weight. Additionally, providing this spaced construction provides better impact protection. Also, as can be appreciated, the flanges **101**, **102**, **103**, **104**, **151**, **152**, **153**, and **154** can serve as grab handles so the laser level tool **10** can be carried or re-positioned.

Additionally, the flanges **101**, **102**, **103**, **104**, **151**, **152**, **153**, and **154** have at least one set of aligned flats which allows the laser level **10** to be accurately re-positioned on its side to project a vertical beam. That is, when sitting upright, the laser level **10** projects a tool in a horizontal plane. The flats allow the laser level **10** to be placed on its sides so that a beam can be projected vertically. The flanges are also designed so as to not interfere with mounting of the tool on a tripod either vertically or horizontally. In the exemplary embodiment, the laser level **10** can be stably positions on a flat surface in at least six orientations. The at least six orientations correspond to the six sides of the cube-shaped control mechanism housing **20**. That is, the laser level **10** can be positioned on a flat surface upright, upside-down or on any of its four sides. In any of these orientations, the laser level **10** will sit stably with only the protective structure **50** resting on the flat surface.

[0072] As shown in, for example, FIG. 3, the upper flanges **101**, **102**, **103**, **104** form a closed shape that is generally square. The structure is open inside the closed square shape, allowing access to the laser projector **30**. Similarly, as is shown in FIG. 8, the lower flanges **151**, **152**, **153**, **154** form a closed shape that is generally square shape. The structure is open inside the closed square shape, allowing access to the bottom surface **22** of the control mechanism housing **20**.

[0073] FIG. 16 shows another exemplary embodiment of a laser level **210**. Unless otherwise stated, the features of this exemplary embodiment are the same as in the previous exemplary embodiment. For example, similar materials can be used in this exemplary embodiment and similar laser mechanisms are also possible. In this instance, the laser projector **230** is disposed at a center of a control mechanism housing **220**. There is a protective structure **250** which includes an upper protective structure **300** and a lower protective structure **350**. The upper protective structure **300** includes flanges **301**, **302**, **303**, **304** which provide a generally square shape. The lower protective structure **350** includes flanges **351**, **352** (and two other flanges not shown) which also provide a generally square shape. In this exemplary embodiment, the upper legs **310** include a single leg **310** at each corner of the control mechanism housing **220**. As can be appreciated, in this embodiment, laser beams projected by the projector **230** do not have to pass through the upper legs **310**.

Accordingly, the upper legs **310** can be made thicker without impeding the projection of a laser

beam. In this exemplary embodiment, the lower legs **360** also comprise a single leg **360** at each corner.

[0074] As shown in FIG. **16**, this exemplary embodiment includes support posts **240**. The support posts **240** connect portions of the control mechanism housing **220**. As can be appreciated, laser beams projected from the laser projector **230** may pass through this section of the housing **220**. Accordingly, having relatively thin supports posts **240** lessens the amount of any disruption of a laser beam.

[0075] Another exemplary embodiment of a laser level is shown in FIG. **17**. As shown in FIG. **17**, the laser level **410** of this exemplary embodiment is generally cylindrically shaped. In particular, the control mechanism housing **420** is roughly cylindrical shaped and the overall laser level **410** is roughly cylindrically shaped. As shown in FIG. **17**, the projector **430** in this embodiment is at a top end of the housing **420**.

[0076] As with the previous embodiments, the laser level **410** of FIG. **17** has a protective structure **450** consisting of an upper protective structure **500** and a lower protective structure **550**. The upper protective structure **500** includes a generally circular flange **501** which is supported by four upper legs **510**. The flange **510** includes a series of bumpers **511**. In the exemplary embodiment, there are four bumpers **511** which are disposed at each of the legs **510**. In other embodiments, there may be a greater number of bumpers **511** or fewer bumpers **511**.

[0077] The lower protective structure **550** likewise includes a generally circular flange **551** which is supported by four lower legs **560**. The lower protective structure **550** also includes a series of bumpers **561**. The bumpers **511**, **561** make it so that if the laser level **410** is placed on its side on a flat surface, the protective structure **450** contacts the flat surface and the control mechanism housing **420** does not. In this exemplary embodiment, because of the shape of the laser level **410**, the laser level **410** can be rolled when on its side if pushed by a user. As with other embodiments, it can also be placed upside down. The bumpers **511**, **561** also allow it to rest on each of its sides as each bumper will resist further rolling.

[0078] In other embodiments, the shape of the laser level could be different. For example, in FIG. **17**, the flanges **501**, **551** are generally circular and the control mechanism housing **420** is generally cylindrical. The shape could be modified so that the flanges are generally elliptical and the control mechanism housing is a generally elliptical cylinder. The major axis of the elliptical shape can be different in different embodiments.

[0079] Simplified schematics for operation of a laser level are shown in FIGS. **18A** and **18B**. As shown in FIG. **18A**, battery pack **200** provides power to a light source **201**, a controller **202** and motor **203**. The light source **201** may be, for example, a laser diode. The controller **202** may, for example, a microcontroller or microprocessor. As shown in FIG. **18B**, there is a light source **201**. The light source **201** projects light through a collimator lens **204** and the light then travels through a prism **205** and finally is reflected off of mirror **206** disposed in the projector **30**. Motor **203** rotates the projector **30**. Alternatively, the motor **203** may rotate the mirror **206** directly. As previously discussed other mechanisms for the laser are possible and part of the present application.

[0080] In an exemplary embodiment, the upper protective structure **100** may be made of a material having a Shore A hardness of 40 to 100; 50 to 100; 60 to 100; 70 to 100; 70 to 90; 60 to 90; 50 to 90; 40 to 90 or 40 to 80.

[0081] In an exemplary embodiment, the lower protective structure may be made of a material having a Shore A hardness of 40 to 100; 50 to 100; 60 to 100; 70 to 100; 70 to 90; 60 to 90; 50 to 90; 40 to 90 or 40 to 80.

[0082] In an exemplary embodiment, the flanges of the upper protective structure may be made of a material having a Shore A hardness of 40 to 100; 50 to 100; 60 to 100; 70 to 100; 70 to 90; 60 to 90; 50 to 90; 40 to 90 or 40 to 80.

[0083] In an exemplary embodiment, the flanges of the lower protective structure may be made of

a material having a Shore A hardness of 40 to 100; 50 to 100; 60 to 100; 70 to 100; 70 to 90; 60 to 90; 50 to 90; 40 to 90 or 40 to 80.

[0084] The legs of the upper protective structure may be made of a material having a higher hardness than the material of the flanges of the upper protective structure. The legs of the upper protective structure may be made of a material having a lower hardness than the material of the flanges of the upper protective structure.

[0085] The legs of the lower protective structure may be made of a material having a higher hardness than the material of the flanges of the lower protective structure. The legs of the lower protective structure may be made of a material having a lower hardness than the material of the flanges of the lower protective structure.

[0086] In another embodiment, the flanges of the protective structure can form a triangular shape and the control mechanism housing can be generally shaped as a triangular prism. In other embodiments, the flanges could form a shape with more sides such as 5 sides (pentagon), six sides (hexagon), seven sides (heptagon), eight sides (octagon), etc. and the control mechanism housing can be shaped with a corresponding structure (i.e., having a cross-section that corresponds to the shape formed by the flanges).

[0087] In other embodiments, the shape formed by the flanges and the control mechanism housing may not correspond. For example, the flanges may form a hexagon shape while the control mechanism housing is generally cube shaped.

[0088] Various different features have been shown and described with respect to different embodiments. It is contemplated that the features of the embodiments could be combined or used in other embodiments. For example, a centrally located projector as shown in FIG. 16 could also be used with a cylindrically laser level of the type shown in FIG. 17.

[0089] While the invention has been described by way of exemplary embodiments, it is understood that the words which have been used herein are words of description, rather than words of limitation. Changes may be made within the purview of the appended claims, without departing from the scope and spirit of the invention in its broader aspects.

Claims

1. A laser level comprising: a mechanism housing; a motor in the mechanism housing; a laser projector at a top of the mechanism housing and configured to project a 360 degree rotary projection laser beam; a battery pack receptacle on the mechanism housing, the battery pack receptacle configured to receive a battery pack such that at least a majority portion of the battery pack is outside of the mechanism housing; a first protective flange spaced apart from the mechanism housing; a second protective flange spaced apart from the mechanism housing; a third protective flange spaced apart from the mechanism housing; and a fourth protective flange spaced apart from the mechanism housing; wherein the battery pack receptacle is disposed between the first protective flange and the protective second flange; and wherein the first flange and the second flange include a material having a Shore A hardness in a range of 40 to 100.
2. The laser level of claim 1, wherein the laser level is configured to be positioned on its side to project the 360 degree rotary projection laser beam as a vertical beam.
3. The laser level of claim 1, further comprising a controller in the mechanism housing.
4. The laser level of claim 3, further comprising a collimating lens in the mechanism housing.
5. The laser level of claim 1 wherein the material has a Shore A hardness in a range of 60 to 90.
6. A laser level comprising: a mechanism housing; a motor in the mechanism housing; a laser projector at a top of the mechanism housing and configured to project a 360 degree rotary projection laser beam; a battery pack receptacle on the mechanism housing, the battery pack receptacle configured to receive a battery pack such that at least a majority portion of the battery pack is outside of the mechanism housing; a first protective flange spaced apart from the

mechanism housing; a second protective flange spaced apart from the mechanism housing; a third protective flange spaced apart from the mechanism housing; and a fourth protective flange spaced apart from the mechanism housing; wherein the first protective flange is parallel to the second protective flange; wherein the second protective flange is parallel to the third protective flange; wherein the third protective flange is parallel to the fourth protective flange; and wherein the battery pack receptacle is disposed between the first protective flange and the protective second flange.

7. The laser level of claim 6, wherein the laser level is configured to be positioned on its side to project the 360 degree rotary projection laser beam as a vertical beam.
 8. The laser level of claim 6, further comprising a controller in the mechanism housing.
 9. The laser level of claim 8, further comprising a collimating lens in the mechanism housing.
 10. The laser level of claim 9, wherein the material has a Shore A hardness in a range of 60 to 90.
 11. A laser level comprising: a mechanism housing; a motor in the mechanism housing; a laser projector at a top of the mechanism housing and configured to project a 360 degree rotary projection laser beam; a battery pack receptacle on the mechanism housing, the battery pack receptacle configured to receive a battery pack such that at least a majority portion of the battery pack is outside of the mechanism housing; a first flange spaced apart from the mechanism housing; and a second flange spaced apart from the mechanism housing; wherein the battery pack receptacle is disposed between the first flange and the second flange; and wherein the first flange and the second flange include a material having a Shore A hardness in the range of 50 to 100.
 12. The laser level of claim 11, wherein the first flange is parallel to the second flange.
 13. The laser level of claim 11, wherein the mechanism housing has a substantially cubical shape.
 14. The laser level of claim 11, further comprising a third flange spaced apart from the mechanism housing; and a fourth flange spaced apart from the mechanism housing.
 15. The laser level of claim 14, wherein the third flange is parallel to the fourth flange.
 16. The laser level of claim 15, wherein the first flange is spaced apart from the mechanism housing by a first distance; wherein the fourth flange is spaced apart from the mechanism housing by a second distance; wherein the first distance is greater than the second distance.
 17. The laser level of claim 16, wherein the laser level is configured to be positioned on its side to project the 360 degree rotary projection laser beam as a vertical beam.
 18. The laser level of claim 16, further comprising a controller in the mechanism housing.
 19. The laser level of claim 18, further comprising a collimating lens in the mechanism housing.
 20. The laser level of claim 19, wherein the mechanism housing has a substantially cubical shape.
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